

Evaluation of Two LiDAR Sensors

Accuracy, precision, hypothesis testing, and simulation

Aurlun Nelson Daniel Kim

STAT 4714 – Professor Courtney Kyger

Fall 2026

Problem and data

Engineering objective

- ▶ Select a LiDAR sensor for distance measurement on an autonomous vehicle.
- ▶ Evaluate measurements collected 5, 10, and 15 ft from a wall.
- ▶ Account for both accuracy and precision.

Safety event: both sensors report a distance at least 0.5 ft shorter than the true distance.

Dataset

Two sensors

Three stopping distances

25 measurements per group

150 measurements total

Statistical methods and assumptions

Accuracy tests

$$H_0 : \mu = d_{\text{true}} \quad H_a : \mu \neq d_{\text{true}}$$

- ▶ Two-sided one-sample t-test
- ▶ Significance level $\alpha = 0.05$
- ▶ $n = 25$ and $df = 24$ for each test

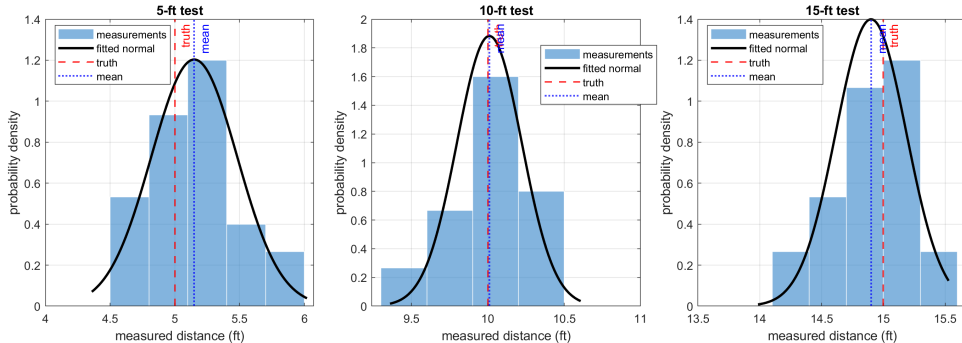
Five groups did not reject normality. Sensor 2 at 10 ft rejected it ($p = 0.0089$), so we interpret that t-test cautiously.

Assumptions

- ▶ Distance is quantitative and continuous.
- ▶ The experimental design supports independent measurements.
- ▶ Lilliefors tests assess normality in each group.

Sensor 1 measurement distributions

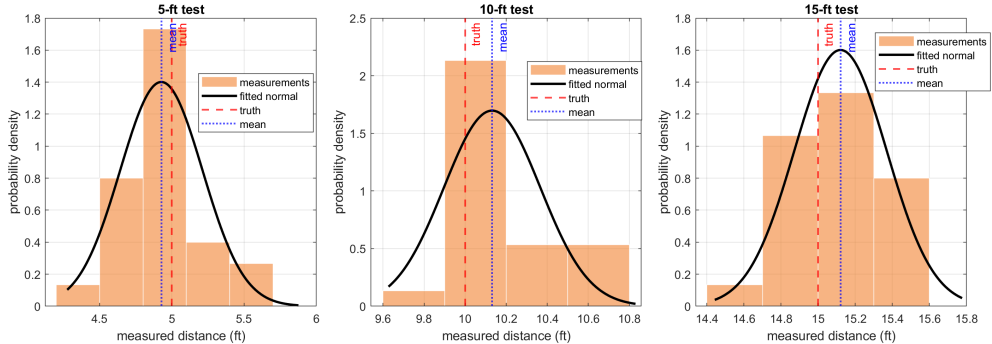
sensor 1: measurements with fitted normal distributions



Red lines show the true distances. Blue lines show the sample means.

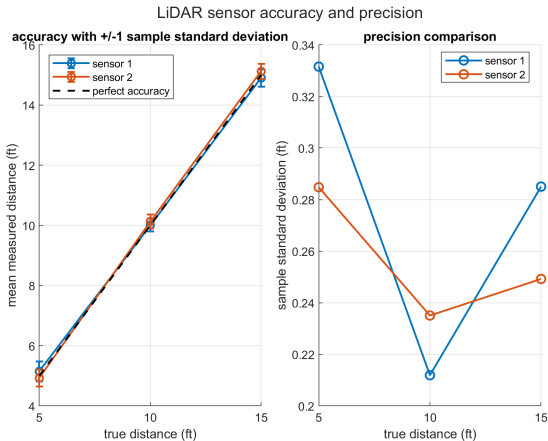
Sensor 2 measurement distributions

sensor 2: measurements with fitted normal distributions



Sensor 2 at 10 ft shows the clearest departure from normality.

Accuracy and precision comparison



Closest mean

- ▶ 5 ft: Sensor 2
- ▶ 10 ft: Sensor 1
- ▶ 15 ft: Sensor 1

Smaller standard deviation

- ▶ 5 ft: Sensor 2
- ▶ 10 ft: Sensor 1
- ▶ 15 ft: Sensor 2

Descriptive statistics

Sensor	Distance	Mean	Std. dev.	Bias
1	5 ft	5.1496	0.3316	+0.1496
1	10 ft	10.0097	0.2119	+0.0097
1	15 ft	14.9006	0.2850	-0.0994
2	5 ft	4.9271	0.2848	-0.0729
2	10 ft	10.1314	0.2350	+0.1314
2	15 ft	15.1208	0.2492	+0.1208

Interpretation

Sensor 1 has the smaller absolute bias at 10 and 15 ft. Sensor 2 has the smaller absolute bias at 5 ft.

One-sample t-test results

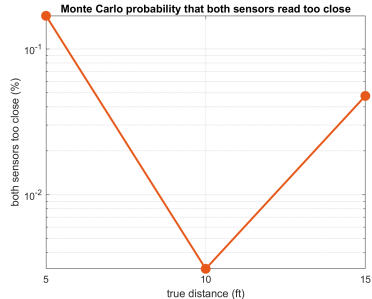
Sensor	Distance	t	p	Decision at $\alpha = 0.05$
1	5 ft	2.2564	0.0334	Reject H_0
1	10 ft	0.2282	0.8214	Fail to reject H_0
1	15 ft	-1.7437	0.0940	Fail to reject H_0
2	5 ft	-1.2803	0.2127	Fail to reject H_0
2	10 ft	2.7960	0.0100	Reject H_0
2	15 ft	2.4231	0.0233	Reject H_0

Sensor 1 differs significantly from truth at one distance. Sensor 2 differs significantly at two distances.

Monte Carlo simulation

For each distance, the simulation generated 1,000,000 independent normal readings using each sensor's sample mean and standard deviation. A trial counted when both readings were at least 0.5 ft too short.

Distance	Both too close
5 ft	0.1687%
10 ft	0.0031%
15 ft	0.0476%



Conclusions and recommendation

Findings

- ▶ Sensor 1 matches the true distance at 10 and 15 ft according to the t-tests.
- ▶ Sensor 2 matches the true distance only at 5 ft.
- ▶ Both-sensor early-reading events remain below 0.17% at every distance.

Recommendation

Select **Sensor 1** for the autonomous vehicle. It provides better accuracy across the tested range and passes two of the three accuracy tests.

Sensor 2 is slightly more precise at 5 and 15 ft, but overestimates distance at 10 and 15 ft.